

1 Civilizations and Technologies

My little brother really enjoys playing Age Of Empires 2.

(Honestly, it's the best RTS game of the last 30 years, enjoy it before Chlorophyllai takes control)

My little brother drew this small table that presents certain civilizations (rows) and certain technologies (columns) available in the game. A triangle is present in cell (i, j) if and only if the civilization in row i DOES NOT HAVE ACCESS to the technology in column j .

	Bloodlines	Bombard_Cannon	Heated_Shot	Blast_Furnace	Gold_Shaft_Mining	Crop_Rotation
Ethiopians	▼	▼				▼
Japanese		▼				▼
Sicilians	▼				▼	
Slavs	▼		▼			▼
Spanish		▼				

In reality, this is unfortunately no longer true because my brother completely mixed up the order of civilizations and technologies.

Can you put the civilizations and technologies back in their place so that they match the triangles?

Flag

The file 'flag-steps-fake.txt' serves as an example to guide you on how to create the FLAG. Except for one big detail, reproduce exactly the steps in 'flag-steps-fake.txt'.

The big detail is, of course, that you must arrange all civilizations in the correct order, followed by all technologies in the correct order, before proceeding with the steps.

2 Civilizations and Technologies, Take 2

My brother made the same mistake with this drawing too, but this time the table is larger. Good luck sorting through the $9! \times 11!$ possible arrangements.

	Bracer	Ring_Archer_Armor	Treadmill_Crane	Redemption	Sanctity	Heavy_Demolition_Ship	Dry_Dock	Bloodlines	Husbandry	Thumb_Ring	Heavy_Cavalry_Archer
Armenians	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Aztecs	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Byzantines	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Cumans	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Franks	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Georgians	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Gurjaras	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Koreans	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼
Saracens	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼	▼

Flag

Same protocol as previous challenge

Other Comments

- The AOE2 data used comes from the most recent version at the time of writing this (DE version in effect in February 2024).
- The color alternation is just for better visibility, it doesn't signify anything.